World Register of Dams

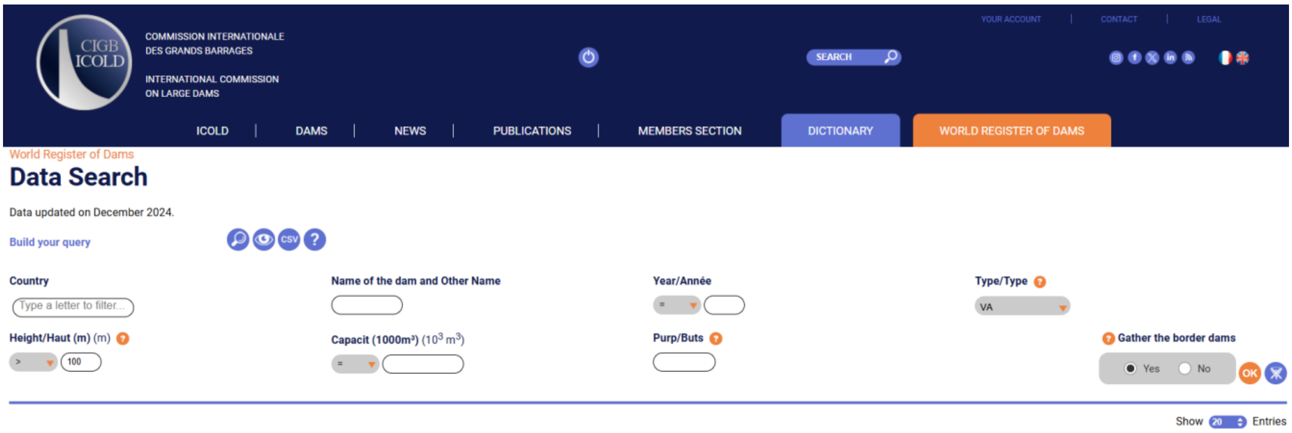
User guide

Interface overview

The search engine interface is divided into two sections.

The upper part, entitled “Build your query”, displays the search criteria, which can be selected and modified. By default, the following criteria are used: Country, Dam name or other name, Year, Type, Height, Capacity, Purpose.

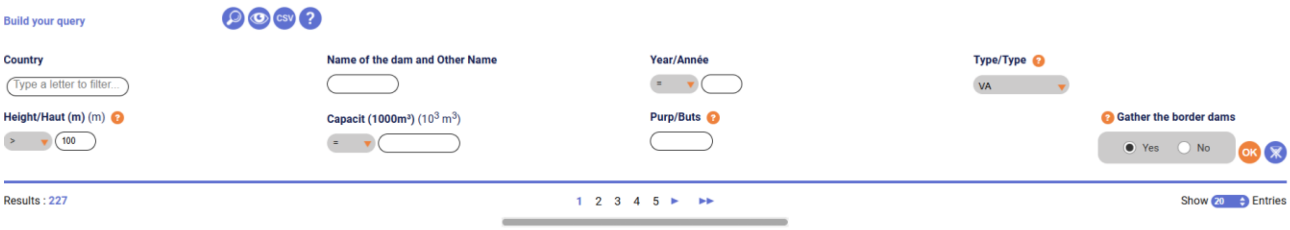
The user specifies the search criteria with the desired values for each criterion, then launches the search by selecting the “OK” button. The search results include dams matching the criteria for all fields with the specified values; for example, if “Type = VA” and “Height>100” are specified, the result lists all dams with an arched cross-section and a height greater than 100 m.



Type =Arch, Height: all the dams with a height above 100 m

The fields are reset by selecting the button. The “Gather the border dams” area allows you to manage the case of international dams by deleting, with the “Yes” option, dams that might be represented twice in the results.

The lower section, entitled “Results”, displays the search results in tabular form. By default, the columns corresponding to the fields indicated above are displayed. Above the table is the number of dams selected.



Country	Dam/Barrage	Year/Année	Type/Type	Height/Haut (m) (m)	Capacit (1000m³) (10³ m³)	Purp/Buts	Other Name/Autre nom	PDF
Argentina	AGUA DEL TORO	1976	VA	120	432,001	CHR		
Argentina	LA VIÑA	1944	VA	106	230,001	SHR	ANTONIO MEDINA ALLENDE,LA VINA	
Australia	GORDON	1974	VA	140	12,400,000	H		
Austria	DROSSEN	1955	VA	112	87,001	H	DROSSEN	
Austria	KOELNBREIN	1977	VA	200	200,001	H		
Austria	KOPS	1965	PG/VA	122	44,601	H	STAUMAUER SPEICHER KOPS	
Austria	SCHLEGEIS	1971	VA	131	129,001	H		
Austria	ZILLERGRUENDL	1985	VA	186	89,901	H		
Bosnia and Herzegovina	GRANCAREVO	1967	VA	123	1,280,000	HICF		
Bulgaria	KARDGALI	1977	VA/PG	104	497,237	HS	KARDJALI	
Bulgaria	TSANKOV KAMAK	2012	VA/PG	131	110,709	HIS		
Chile	RAPEL	1968	VA	112	680,001	H		

Result of a query based on dams with an arched section over 100 m high

The last column (PDF) is used to obtain the complete data sheet for a given dam.

The header line has a dual purpose. It can be used to sort the table, for example to display all dams in descending order of height.

Country	Dam/Barrage	Year/Année	Type/Type	Height/Haut (m) (m)	Capacit (1000m³) (10³ m³)	Purp/Buts	Other Name/Autre nom	PDF
China	JINPING 1	2013	VA	305	7,760,000	HC		
China	XIAOWAN	2012	VA	294	15,043,000	HCIN		
China	BAIHETAN	2021	VA	289	20,627,000	H		
China	XILUODU	2014	VA	286	12,914,000	HCN		
Iran	BAKHTIYARI	2026	VA	275	4,845,000	HC		
Türkiye	YUSUFELI	2022	VA	275	2,130,000	H	YUSUFELI	
Georgia	ENGURI	1987	VA	272	1,100,000	HI	INGURI	
China	WUDONGDE	2021	VA	270	7,408,000	H		
Italy	VAJONT	1960	VA	262	168,731	H		
China	SHANGSAI		VA	250	1,100,000	H		

Result of a query based on arch dams over 100 m high, sorted by decreasing height.

It also allows users to refine their search by typing the values at the head of one or more columns.

Country (Türkiye)	Dam/Barrage	Year/Année	Type/Type	Height/Haut (m) (m)	Capacit (1000m³) (10³ m³)	Purp/Buts	Other Name/Autre nom	PDF
Türkiye	YUSUFELI	2022	VA	275	2,130,000	H	YUSUFELI	
Türkiye	DERINER	2014	VA	249	1,969,000	H	DERINER	
Türkiye	ERMENEK	2009	VA	218	4,583,000	H		
Türkiye	BERKE	2002	VA	201	427,001	H		
Türkiye	OYMAPINAR	1984	VA	185	235,997	H		
Türkiye	KARAKAYA	1987	VA	173	9,580,000	H		
Türkiye	GÖKÇEKAYA	1972	VA	159	910,001	H	GOKCEKAYA	
Türkiye	SIR	1991	VA	116	1,120,000	H		
Türkiye	KEMER	1958	VA	114	544,001	IH		
Türkiye	SÖYLEMEZ	2024	VA	113	1,294,600	IH	SOYLEMEZ	

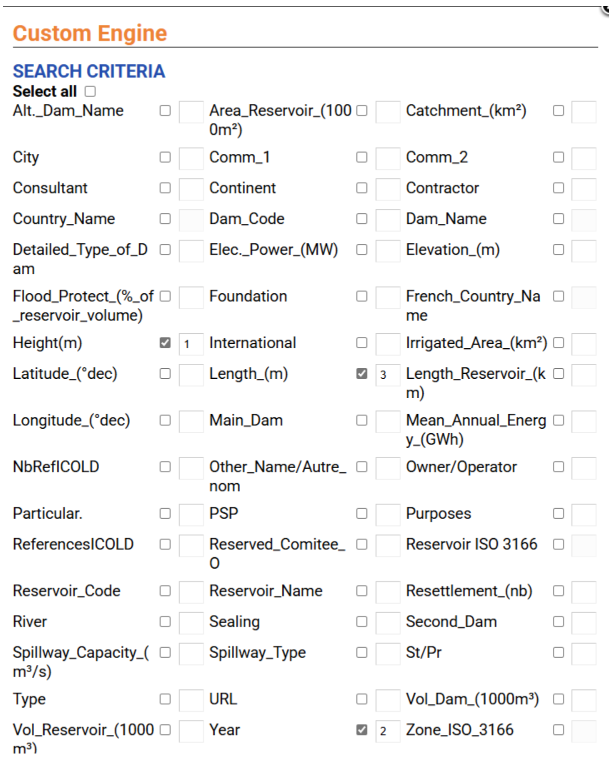
Results of previous research limited to Turkish dams only

The button CSV at the top of the page lets you export the table in CSV format (UTF8, semicolon separator, delimited text).

Search engine customization!

1/ Query definition

To choose the fields to be searched, select the button and tick the search criteria in the desired display order. In the example below, the user requests display of the search fields "Height", "Year" and "Length".

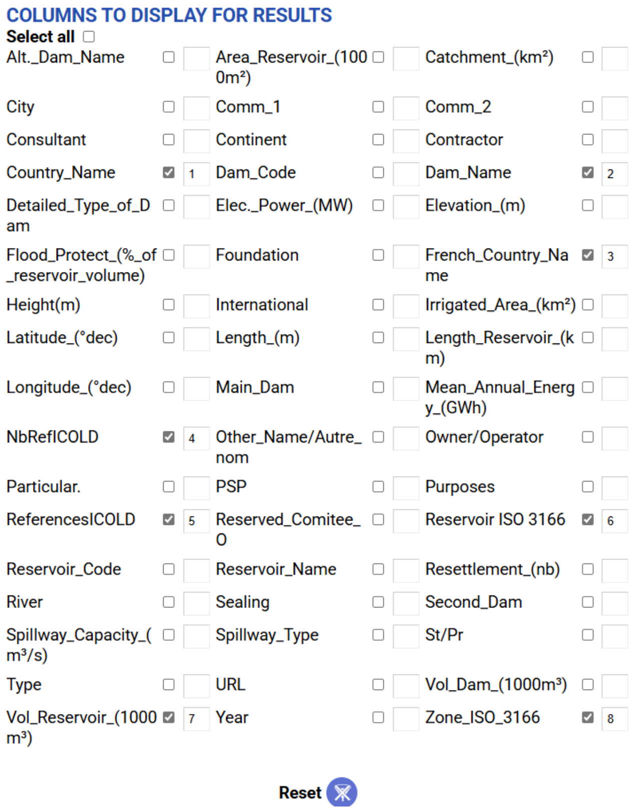


The "Height", "Year" and "Length" fields will be displayed in the "Build your query" section

Field selection is validated by selecting the save button at the bottom of the customization page.

2/ Results table customization

On the lower part of the same screen, you can customize the results table (with fields that may differ from those defining the query).


COLUMNS TO DISPLAY FOR RESULTS

Select all

Alt._Dam_Name		Area_Reservoir_(1000m²)		Catchment_(km²)	
City		Comm_1		Comm_2	
Consultant		Continent		Contractor	
Country_Name	1	Dam_Code		Dam_Name	2
Detailed_Type_of_Dam		Elec._Power_(MW)		Elevation_(m)	
Flood_Protect_(%_of_reservoir_volume)		Foundation		French_Country_Name	3
Height(m)		International		Irrigated_Area_(km²)	
Latitude_(°dec)		Length_(m)		Length_Reservoir_(km)	
Longitude_(°dec)		Main_Dam		Mean_Annual_Energy_(GWh)	
NbRefICOLD	4	Other_Name/Autre_nom		Owner/Operator	
Particular.		PSP		Purposes	
ReferencesICOLD	5	Reserved_Comitee_O		Reservoir ISO 3166	6
Reservoir_Code		Reservoir_Name		Resettlement_(nb)	
River		Sealing		Second_Dam	
Spillway_Capacity_(m³/s)		Spillway_Type		St/Pr	
Type		URL		Vol_Dam_(1000m³)	
Vol_Reservoir_(1000m³)	7	Year		Zone_ISO_3166	8

Reset

Selection and classification of results table fields

The button at the bottom of the screen also validates this section.

Definition of database fields

The database contains the fields shown in the table below:

Item	Definition
Continent	Continent, distinguishing, for America, North, South or Central America
Country Name	Short name of the country represented at the UN. For international dams, country names are separated by the character "/"
French Country Name	Short name in French of the country represented at the UN. For international dams, country names are separated by the character "/"
Dam Name	Dam name
Second Dam	The field contains the letter “S” if there is another large dam around the same reservoir, called the “main dam”. There may be several secondary dams around the same reservoir, but only one main dam.
MainDam	Name of main dam, if any
Reservoir Name	Reservoir name (often identical to dam name)

Year	Year of commissioning. For dams under construction, projected year of commissioning
Particular.	C = dam under construction. L = lowered dam. H = dam raised. R = dam rebuilt.
International	For international dams, name of the country(ies) in charge of the dam.
River	Name of river downstream of the dam
City	Name of nearest town
St/Pr	Name of department, province or state
Type	Dam type: CB = buttress dam, BM = movable dam, ER = rockfill dam, MV = multiple arch dam, PG = gravity dam, TE = earth dam, VA = arch dam, XX = not listed. If a dam has several sections of different types, the types are listed, separated by "/".
Sealing	The type of seal is defined by two letters, the first indicating the position, the second its nature. For position, we indicate: f = upstream, h = homogeneous, i = inner core, x = unknown or not listed. The type is indicated by: a = bituminous, c = concrete, e = earth, m = metal, p = plastic or geomembrane, x = unknown or not listed. If the dam has several sections of different types, the waterproofing types are listed, separated by "/".
Foundation/	One of the following values: R = rock, S = soil, R/S = mixed foundation, X = unknown
Height (m)	Dam height in meters above foundation
Length (m)	Crest length in meters
Vol Dam ($1000m^3$)	Total volume of dam in 1000 m^3
Vol Reservoir ($1000m^3$)	Volume of full reservoir, expressed in thousands of m^3
Area Reservoir ($1000m^2$)	Surface area of full tank, in thousands of m^2
Length Reservoir (km)	Length of full reservoir in km
Purposes	Accepted values are: C = flood control, I = irrigation, H = hydropower, F = fish farming, N = navigation, R = recreation, S = water supply, T = storage of industrial tailings, X = other not listed. In the case of multiple uses, the uses are listed, without separators, starting with the most important
Catchment. (km^2)	Catchment area in km2
Spillway capacity (m^3/s)	Maximum flood discharge
Type Spillway/Type Evac.	Type of spillway: L = free, V = gated, L/V = mixed free and gated, X = not listed, N = no spillway
Owner	Owner's name
Consultant	Name of design office
Contractor	Name of construction company
Comm 1	Free comment
Comm 2	Free comment
URL	URL(s) of an Internet page concerning the dam or its reservoir. Several addresses are separated by the character "|".
PSP	Indicates the upper (UPPER) or lower (LOWER) dams of pumped storage schemes
Elec. Power (MW)	Installed electrical capacity in MW
Mean Annual energy (GWh)	Average annual energy produced in GWh.

Irrigated Aarea (km²)	Irrigated area in km²
Flood Protect. (% of reservoir volume)	Ratio (in percent) of maximum volume stored in reservoir during floods to total reservoir volume
Resettlement (nb)	Number of people displaced for project construction
Sedimentation	Percentage ratio of sediment volume stored in reservoir to total reservoir volume
Detailed Type of dam	Detailed type of boom with the following recommended values: • BM: movable dam • GM: masonry gravity dam • GCC: conventional concrete gravity dam • GHCC: lightened conventional concrete gravity dam • GRCC: roller-compacted concrete gravity dam • GRFC: rockfilled concrete gravity dam • GSSG: cement-sand-gravel gravity dam • GCSGR: cement-sand-gravel-riprap gravity dam • AGCC: conventional concrete arch-gravity dam • AGM: masonry arch-gravity dam • AGRCC: arch-gravity dam in roller-compacted concrete • AM: masonry arch dam • ACC: conventional concrete arch dam • ARCC: roller-compacted concrete arch dam • ARFC: rock-filled concrete arch dam • DSB: flat slab buttress dam • MAB: multiple-arch buttress dam (MV type) • MHB: broad-head buttress dam • TEH: homogeneous earth dam • CFRD: rockfill dam with concrete upstream mask • AFRD: rockfill dam with bituminous upstream mask • AFED: earthen dam with bituminous upstream mask • PTE: earth dam with geomembrane upstream mask • MFED: fill dam with metallic upstream mask • ACED: bituminous central-core dam • CCED: concrete core dam • ECED: zoned clay core dam • FSHD: trapezoidal hard-fill dam • CSG: cement-sand-gravel dam • TDDS: downstream tailings dam • TDC: Centrally placed tailings dam • TDUS: upstream tailings dam If a dam has more than one type of structure, the types are separated by a "/".
Alt Dam Name	Dam alias. In the case of multiple aliases, the different values are separated by a comma
Elevation	Altitude of dam crest
Longitude (°dec)	Longitude (WGS84) of the middle of the dam crest in decimal degrees
Latitude (°dec)	Latitude (WGS84) of the middle of the dam crest in decimal degrees
Dam code	Dam Identification Code
Reservoir Code	Reservoir Identification Code
Zone Iso 3166	ISO 3166 (alpha 2) zone of the dam implantation site. For international dams, the codes for each country are separated by

	the character "/"
Réservoir Iso 3166	ISO 3166 (alpha 2) area of the reservoir. For reservoirs located in several countries, the codes for each region are separated by the character "/"
NbReferencesICOLD	Number of publications in the ICOLD publications database where the dam is cited.
ReferencesICOLD	Citations of articles about the dam in the ICOLD publications database. Optionally separated by the character "|"

Tips and precautions

When using the database, particularly when exporting data, please note the following points:

- The names of dams and reservoirs, towns, rivers, etc., may contain accented letters that are not commonly used in other countries. For dam names, the search is performed without taking accents into account, but this is not the case for other fields.
- Care should be taken when adding reservoir volumes and dam surfaces, and when counting reservoirs if the search result includes both secondary and secondary dams. It may be wise to add the “Second dam” field, and to take the value of this field into account in subsequent syntheses.
- To select dams in a numerically defined geographical area, the best solution is to provide a range of values for longitudes and another for latitudes.